Culturally-Aware Image Captioning for Indigenous Languages of the Americas:
A Retrieval-Grounded and Verification-Oriented Vision-to-Translation Pipeline

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Abstract

Generic vision–language models (VLMs) frequently fail to preserve culturally specific information present in images, creating a bottleneck for multilingual image captioning systems targeting Indigenous languages. In the AmericasNLP 2026 Cultural Image Captioning for Indigenous Languages (CICIL) shared task, captions are generated directly in Indigenous languages of the Americas from culturally situated images. Existing systems typically generate an intermediate Spanish caption with a generic VLM and then translate it using retrieval-augmented language models, but culturally salient visual concepts are often lost before translation begins. We investigate whether retrieval-grounded cultural reasoning can improve this stage.

We present a two-stage captioning pipeline that augments vision processing with cultural retrieval and verification. Stage 1 uses structured cultural visual question answering (VQA), Wikipedia-derived retrieval, and a retrieval-grounded interrogation loop inspired by caption-questioning frameworks. Retrieved cultural hypotheses are verified through closed visual questions before being incorporated into Spanish descriptions. Stage 2 performs culturally indexed retrieval over parallel corpora and translates the Spanish intermediate into the target language using Gemini 2.5 Flash.

Across five Indigenous languages (Guaraní, Bribri, Yucatec Maya, Wixárika, and Nahuatl), retrieval substantially increases explicit cultural references, raising culture-name rates from near-zero to 18–37 of 50 dev captions (29–38 for a 7B teacher). However, corresponding chrF++ improvements are small and inconsistent. Through human evaluation, retrieval audits, decoding ablations, and extensive qualitative analysis, we show that standard reference-based metrics are largely insensitive to cultural correctness, hallucinations, retrieval-induced errors, and faithfulness improvements. We further demonstrate that verification ability is capability-gated: a 2B model fails as a verifier even under closed questioning, while a local 7B model reliably distinguishes true from false claims. Our results suggest that culturally grounded captioning is constrained less by translation quality than by cultural retrieval, verification capacity, and evaluation methodology.

1 Introduction

The AmericasNLP 2026 Cultural Image Captioning for Indigenous Languages (CICIL) shared task (AmericasNLP 2026 Organizers, 2026) asks systems to generate image captions directly in Indigenous languages of the Americas. Unlike conventional image captioning benchmarks, CICIL emphasizes cultural grounding: images depict artifacts, ceremonies, landscapes, and community practices that are often meaningful only within specific cultural contexts.

The strongest systems in recent shared-task settings adopt a two-stage pipeline (Dhawan et al., 2026). A vision–language model first produces an intermediate Spanish description, which is then translated through retrieval-augmented language models into the target language. While effective, this design inherits a fundamental limitation: any cultural information omitted in the Spanish intermediate cannot be recovered downstream.

Our work starts from a simple observation. Culture recognition from pixels alone is an ill-posed inverse problem. Many visual motifs recur across unrelated traditions, geographic regions, and historical periods. Ñandutí lace patterns, for example, descend from broader Tenerife lace traditions and cannot always be uniquely identified from image evidence alone. Cultural attribution therefore requires not only visual evidence but also contextual priors. The CICIL dataset already provides one such prior through image-level cultural metadata, yet most existing pipelines do not incorporate this information into visual reasoning.

We investigate whether retrieval-grounded cultural reasoning can improve culturally situated caption generation. Our initial experiments with cultural VQA prompting showed only marginal gains, motivating a stronger retrieval-based approach. We therefore introduce a retrieval-grounded captioning architecture in which external cultural knowledge is supplied at inference time and verified through visual questioning before entering the final caption.

The paper addresses three research questions:

RQ1: Does culturally grounded Stage-1 processing improve caption quality compared with generic vision–language baselines?

RQ2: How does performance vary across languages with different resource availability?

RQ3: Which categories of cultural information are most difficult to ground reliably?

Our contributions are:

- A retrieval-grounded cultural captioning pipeline combining visual reasoning, cultural retrieval, and verification.
- Evidence that retrieval, rather than model scale alone, is the primary constraint on cultural naming.
- A capability analysis showing that verification emerges only above a minimum model scale.
- An empirical demonstration that chrF++ fails to reflect many important dimensions of cultural caption quality.
- Extensive qualitative analysis of retrieval failures, hallucinations, calibration behavior, and evaluation limitations.

2 Related Work

2.1 Cultural Grounding in Vision–Language Models

Vision–language models inherit cultural biases from predominantly Western web-scale training data. CLIP-style contrastive pretraining (Radford et al., 2021) produces powerful multimodal representations but remains sensitive to geographic and cultural imbalances. Benchmarks such as MaRVL (Liu et al., 2021) demonstrate large performance drops when visual reasoning is evaluated across cultures rather than within familiar English-centric settings.

More recently, CulturalVQA (Nayak et al., 2024) and related benchmarks (Romero et al., 2024) have shown that state-of-the-art systems still struggle with culturally specific visual concepts. Fine-tuning approaches such as CultureCLIP (Huang et al., 2025) improve cultural recognition through synthetic contrastive examples, but such methods require large curated image collections. Under CICIL's small-data constraints, prompting and retrieval become more practical alternatives.

The CIC framework (Yun and Kim, 2024) introduced a VQA-before-captioning paradigm in which a model first identifies cultural elements and then synthesizes a caption. Our work adopts this idea but combines it with retrieval-grounded reasoning and downstream translation.

2.2 Cultural Retrieval

Retrieval-augmented systems have become increasingly important for low-resource NLP (Lewis et al., 2020). Instead of requiring all knowledge to be encoded in model parameters, retrieval supplies relevant information dynamically at inference time.

For cultural grounding, this distinction is particularly important. In our experiments, even a 7B teacher model rarely names cultures unaided, while retrieval dramatically increases culture naming rates. This suggests that cultural knowledge is not merely missing from smaller models but is insufficiently accessible even in larger ones.

A key insight from our findings is that retrieval quality and retrieval usage are distinct. We observed examples where retrieval correctly retrieved relevant cultural material but the model copied irrelevant metadata instead of grounding itself in the image.

2.3 Interrogation and Verification

Our retrieval-grounded interrogation loop follows the ChatCaptioner family of approaches (Zhu et al., 2023), where an LLM asks questions and a vision model answers them before caption synthesis. Related work includes Video ChatCaptioner (Chen et al., 2023), IdealGPT (You et al., 2023), and visually grounded dialogue systems such as Visual Dialog (Das et al., 2017) and GuessWhat?! (de Vries et al., 2017).

The verification component is closely related to Chain-of-Verification (Dhuliawala et al., 2024) and Woodpecker (Yin et al., 2024). However, unlike prior work, we explicitly measure *when* verification works.

Language	Dev	Caption len.	Bank pairs
Guaraní	50	21.6	15,494
Bribri	50	14.9	8,297
Yucatec Maya	50	11.3	14,332
Nahuatl	50	8.3	16,119
Wixárika	50†	17.0	9,940

Table 1: Development data and Stage-2 retrieval banks. "Caption len." = mean gold target-caption length (words). †Plus a 20-image pilot set with gold Spanish captions. Banks are built from AmericasNLP shared-task corpora (Mager et al., 2021; Ebrahimi et al., 2023), the Wixárika–Spanish corpus (Mager et al., 2018), and the YUA-ES Communicative Contexts Corpus (Molina-Villegas et al., 2026).

A central finding of this paper is that verification itself is capability-gated. A 2B model answered "yes" to most verification prompts regardless of correctness, while a local 7B model reliably rejected false claims. Verification therefore appears to require a minimum level of visual reasoning capacity before verification-based pipelines become effective. Notably, ChatCaptioner's own error analysis already attributes 94% of its incorrect captions to wrong answers from the answerer (Zhu et al., 2023); our results characterize where that answerer capacity turns on.

3 Data

The CICIL development data are extremely small: each of the five languages provides 50 development images. Wixárika additionally provides a 20-image pilot set that is uniquely valuable because it includes gold Spanish captions (mean 13.7 words), enabling direct evaluation of Stage 1.

Resource availability varies dramatically across languages (Table 1), creating a natural testbed for understanding how retrieval affects low-resource systems.

4 Method

4.1 Stage 1: Cultural VQA

The original Stage 1 system decomposes image understanding into four cultural categories: (i) ceremony and ritual, (ii) material culture, (iii) landscape and geography, and (iv) kinship and community. A small VLM (SmolVLM-2B; Marafioti et al., 2025) answers category-specific questions and then synthesizes a Spanish description from the answers, following the VQA-before-captioning paradigm (Yun and Kim, 2024).

Our initial experiments revealed that verbose cultural explanations lower chrF++ because references are short. A constrained single-sentence synthesis restored performance while preserving cultural reasoning.

4.2 Retrieval-Grounded Cultural Context

We augment Stage 1 with two cultural retrieval channels.

Image retrieval. We construct SigLIP (Zhai et al., 2023) embedding indices over culture-specific Wikimedia Commons image collections. Retrieved neighbors provide visual examples associated with the target culture.

Text retrieval. We build culture-specific Wikipedia knowledge banks and retrieve relevant passages using multilingual MiniLM embeddings (Reimers and Gurevych, 2020). These passages serve as cultural context during synthesis.

Retrieved evidence is associated with confidence bands: *strong match* ($\geq$ 0.80 similarity) and *possible match* (0.55–0.80). Only strong evidence supports direct assertions; possible matches are hedged explicitly.

4.3 Multi-Agent Interrogation

The final version of Stage 1 adopts a retrieval-grounded interrogation architecture:

1. Generate a base caption.
2. Retrieve candidate cultural concepts.
3. Generate closed verification questions.
4. Answer questions using a VLM.
5. Assemble a final caption from verified evidence.

The questioner and assembler are text-only LLMs; the answerer is a local 7B VLM (Qwen2.5-VL; Qwen Team, 2025). This design transforms open cultural speculation into a sequence of verifiable claims.

4.4 Capability-Gated Verification

Verification performance differed dramatically by model size.

The 2B model displayed extreme yes-bias, producing 82 affirmative responses out of 82 verification attempts. On planted decoys, it accepted clearly false claims and sometimes contradicted its own rationale.

In contrast, a local 7B model correctly rejected all negative controls and successfully verified true positives.

These results indicate that visual verification appears only above a capability threshold.

4.5 Stage 2 Translation

The final Spanish description is translated using retrieval-augmented Gemini 2.5 Flash (Comanici et al., 2025).

We index parallel corpora using multilingual MiniLM embeddings and FAISS (Johnson et al., 2019) and retrieve top-k Spanish–Indigenous examples. Retrieved pairs are supplied as demonstrations during translation.

Temperature 0.7 decoding was adopted after ablation experiments revealed severe degeneration under greedy decoding.

5 Results

5.1 Stage 1 Evaluation

Using the Wixárika pilot's gold Spanish captions, we score Stage-1 Spanish intermediates with chrF++ (Popović, 2017):

Configuration	chrF++
Generic	21.0
Cultural VQA (verbose)	16.2
Cultural VQA (concise)	20.9

Table 2: Spanish-side proxy evaluation (Wixárika pilot, n=20).

Concise cultural prompting reaches parity with generic captioning while reducing internal inconsistencies.

5.2 Human Evaluation

Single-annotator results over 15 images (0–2 rubric per dimension), judged on English pivot translations of the Spanish captions (Hendy et al., 2023; Kocmi et al., 2023):

The main conclusion is that cultural prompting alone generates slightly more cultural content but remains largely indistinguishable from generic captioning.

5.3 Decoding Ablation

Greedy decoding created severe repetition loops:

Dimension	Generic	Cultural
Cultural accuracy	0.00	0.13
Faithfulness	1.73	1.53
Fluency	1.93	1.87

Table 3: Human evaluation of Stage-1 arms. Preference outcomes: tie 60%, cultural preferred 27%, generic preferred 13%.

Language	Degen. before	Degen. after
Bribri	64%	28%
Wixárika	86%	30%

Table 4: Degenerate-caption rate under greedy decoding (before) vs temperature 0.7 with fixed seed (after), 50 dev images per language.

Switching to temperature-based sampling reduced degeneration dramatically and improved chrF++ without harming stronger languages.

5.4 Retrieval Results

Retrieval dramatically increases explicit cultural references (Table 5).

Retrieval therefore succeeds at increasing cultural specificity even when chrF++ remains nearly unchanged.

5.5 End-to-End Task Results

Table 6 reports end-to-end chrF++ against the target-language references (50 dev images per language; Stage-2 translation with k=5 retrieved demonstrations, temperature 0.7, fixed seed).

The pattern is heterogeneous. The 7B teacher with retrieval gains substantially on Guaraní (+1.69) and Wixárika (+1.85); the retrieval student gains most where resources are thinnest (Bribri +1.19, the largest student-side gain); and Maya and Nahuatl lose about one point under retrieval—exactly the two languages where the concept rate stayed at 0–1 of 50 (Table 5), so retrieval added noise without a concept payoff. Set against the order-of-magnitude changes in cultural specificity, these movements are small: the metric and the behavior have largely decoupled.

5.6 Capability Ladder

Wixárika pilot results across system versions:

The remaining gap to frontier performance is roughly 1–1.5 chrF++.

Language	No RAG	RAG student	RAG teacher
Culture-name rate (of 50)			
Guaraní	3	37	37
Wixárika	0	21	38
Maya	1	32	38
Nahuatl	1	18	29
Bribri	0	34	37
Concept rate (of 50)			
Wixárika	0	24	
Bribri	5	25	
Guaraní	1	5	
Nahuatl	0	1	
Maya	0	0	

Table 5: Cultural specificity with and without retrieval. "Culture-name rate" counts captions naming the target culture; "concept rate" counts captions naming a specific cultural concept (artifact, dance, site).

Language	Generic	+RAG	Distill	Teacher
Guaraní	19.35	19.44	19.31	**21.04**
Wixárika	13.18	13.34	13.17	**15.03**
Maya	**19.48**	18.52	18.50	17.65
Nahuatl	**15.89**	15.34	15.05	14.92
Bribri	6.81	**8.00**	7.31	7.07

Table 6: End-to-end chrF++ on dev (target language, n=50 per language). Generic = SmolVLM-2B without retrieval; +RAG = the same student with retrieval; Distill = the RAG-distilled student; Teacher = 7B Qwen2.5-VL with retrieval.

6 Analysis

RQ1: Does cultural grounding help? Cultural prompting alone does not improve end-to-end performance. Retrieval substantially increases cultural specificity (Table 5) but produces limited and language-dependent metric gains (Table 6). Human evaluation suggests that cultural prompting yields marginally richer cultural descriptions at the cost of slight reductions in faithfulness.

RQ2: Resource effects. Resource disparities appear throughout the pipeline. Bribri has the smallest retrieval resources and remains difficult across all systems, yet it shows the largest student-side end-to-end gain under retrieval (+1.19 chrF++) alongside some of the largest gains in cultural concept rates—inverting the naive expectation that retrieval helps most where data is plentiful. Low-resource settings may benefit disproportionately when even modest external knowledge becomes available.

System	chrF++
SmolVLM-RAG	18.44
RAG distill	18.79
Verify-RAG*	20.50
Local 7B base	21.47–21.75
Agent v4.2	21.73
Agent v4.3.1 (local)	21.68
Agent final (strict agg., local)	21.05
Gemini direct	22.71

Table 7: Capability ladder on the Wixárika pilot (chrF++ vs gold Spanish). *The Verify-RAG score is inflated for the wrong reason: its 2B verifier accepted every claim (§4.4), so its gain reflects longer, more assertive captions, not verification. "Agent final" applies the strictest aggregation policy (retrieval-whitelisted naming, decorative-flourish removal); its drop below v4.3.1 is largely a metric artifact—removing decorative mentions of the culture also removes tokens shared with references that name it.

RQ3: Which cultural concepts ground reliably? Artifacts and distinctive cultural objects appear most retrievable. Landscape concepts proved much harder: several initially promising examples collapsed under manual audit because retrieval supplied plausible but unsupported regional labels. This finding motivated stricter auditing procedures and highlights the danger of equating concept naming with genuine grounding.

What the metric cannot see. Across the interrogation arm's development we closed six distinct faithfulness failure classes—each confirmed by manual audit against the images—and chrF++ moved by less than half a point in either direction at every step (Table 7). The converse also holds: the Verify-RAG configuration gained two chrF++ points from a 2B verifier that we later showed accepts every claim it is asked (§4.4), and degenerate repetition loops still scored nontrivially. The effect can even invert: enforcing the no-flourish rule (removing decorative clauses like "evoking Wixárika culture") *costs* about 0.6 chrF++, because the references legitimately name the culture—the metric rewards keeping unfounded decoration. On this benchmark, reference overlap and caption faithfulness have effectively decoupled; we therefore treat chrF++ as a sanity check rather than an objective, and rest the system comparison on specificity rates, audits, and human evaluation.

Failure modes found only by auditing. Manual inspection of full interrogation transcripts surfaced failure classes that no aggregate metric flagged. Verification questions that presuppose a base-caption detail can launder that detail into a "verified" claim (the answerer confirms the presupposition rather than checking it); one-word verdicts can contradict their own rationales, and assembly that trusts verdicts over rationales inherits the error; and the answer channel can inject fabricated cultural vocabulary that retrieval never supplied—in one out-of-domain probe the answerer invented a garment name that does not exist, which the assembler then asserted as fact. Each class was fixed structurally (question phrasing constraints, rationale-over-verdict aggregation, and restricting cultural naming to retrieved vocabulary) rather than by adding prompt instructions, which we found do not reliably bind. One residual remains: when the same model writes the base caption and answers the verification questions, its perceptual errors self-confirm. ChatCaptioner's error analysis attributes 94% of its incorrect captions to the answerer (Zhu et al., 2023); our audits show the failure persists even with verification in the loop, and independent witnesses are the natural next step.

Case studies (Figure 1). Three images make the preceding arguments concrete. In Figure 1a, the reference expects *jaripeo* — event knowledge no image-only system can recover — while the retrieval-only baseline invents a different error entirely, relocating the corral to "Wirikuta, San Luis Potosí"; the interrogation arm's caption ("a group of animals, probably cows or oxen, in front of a wooden structure with a metal roof") is the honest ceiling. Figure 1b shows the three knowledge channels pulling in different directions on one image: the generic model's parametric prior misattributes the dress as *Mexican*; retrieval supplies the correct garment vocabulary (a *typói* with *ñandutí* lace) but drags in an invented location (Itauguá, Paraguay — the ñandutí weaving center, copied from the retrieved context rather than the image); and the interrogation arm grounds the event from the poster's legible text ("Mundial de Chamamé... Corrientes, Argentina"). Two residual errors survive even there. The year is misread (2013 for the poster's 2012), and a phantom "fringed shawl" persists although the direct action probe answered correctly ("holding the edge of her dress") — the second witness hallucinated the shawl, the questioner asked about it *as a presupposition* ("as described in the second description..."), violating its own neutral-phrasing rule, the answerer echo-confirmed it across three follow-ups, and at assembly four confident affirmations outvoted the one correct direct observation. The transcript compresses our two structural findings into one image: prompt rules do not reliably bind, and confidence-weighted aggregation amplifies presupposition echoes. In Figure 1c the baseline turns a suspension footbridge into "a central cobblestone arch, a Wixárika cultural and artisanal element" in Wirikuta; the interrogation arm simply describes the bridge, the river, and the terrain. Across all three, chrF++ moves by fractions of a point.

Gold captions describe more than the image. Reading the 20 Wixárika pilot references against their images, roughly two-thirds contain community knowledge that is not visually discernible: event context ("bulls resting *before the jaripeo begins*"), purpose ("degraining maize *to make nixtamal*"), destination and cause ("taking her daughter *to school* because the river has risen"), or encyclopedic usage notes. Not all of this gap is unbridgeable, however. One reference names the image's stilt structure a *carretón* — a term we initially suspected was an error (its standard sense is "cart"), but which the ethnographic literature attests for exactly this Wixárika structure: a raised, thatched-roof granary on wooden posts, used to store maize and as sleeping quarters.[1] The concept is fully visible in the image and publicly documented, yet absent from our Wikipedia-derived banks — a bank *coverage* ceiling rather than a perception limit, and a concrete argument for harvesting retrieval banks from ethnographic sources. This has two further consequences beyond the obvious ceiling on image-only systems. First, the reference style rewards *fabricating* plausible context: a baseline that invents a plausible region name can gain overlap where honest silence cannot, so the metric points away from faithfulness. Second, it reframes the task itself: the gold captions record what the community knows about a scene, not what the scene shows—and the principled way to close that gap is to supply real context (retrieval, metadata, community input) rather than to ask a vision model to guess it.

[1] Described in ethnographic work on Wixárika communities of southern Durango; see e.g. https://www.thefreelibrary.com/Parentesco+y+relaciones+de+genero+en+una+localidad+Wixarika+(Huichol)...-a0350783750.

(a) Context beyond pixels (b) Three knowledge channels (c) Invented location

Figure 1: Case studies (§6). (a) The reference caption reads "bucking bulls resting *before the jaripeo begins*" — nothing in the frame identifies a rodeo, and the retrieval-only baseline relocates the scene to the sacred site Wirikuta. (b) A Paraguayan folk dress at an Argentine chamamé festival: the generic model calls the dress *Mexican*, the retrieval student names the garment correctly (*typói*, *ñandutí*) but invents a location, and the interrogation arm grounds the event from the poster text. (c) The baseline describes "a cobblestone arch in the Wirikuta region, San Luis Potosí"; the interrogation arm describes the suspension footbridge that is actually there. Images: AmericasNLP 2026 CICIL dataset (CC BY-NC 4.0). All translations ours.

7 Conclusion

We presented a retrieval-grounded cultural image captioning pipeline for Indigenous languages of the Americas. Cultural prompting alone provided minimal gains, but retrieval substantially improved cultural specificity. At the same time, our experiments reveal that cultural captioning is ultimately constrained not by translation but by retrieval quality, verification capability, and evaluation methodology.

Most importantly, we show that standard reference-based metrics fail to capture many of the qualities that matter in culturally situated captioning systems. Future work should therefore prioritize verified cultural grounding, community-informed evaluation, and retrieval methods that provide trustworthy contextual knowledge rather than encouraging plausible cultural guessing.

Limitations

This work has several limitations.

First, reference-based metrics are largely insensitive to cultural correctness. Multiple faithfulness improvements, hallucination fixes, and verification gains produced almost no change in chrF++. Conversely, degenerate repetition loops still received nontrivial scores.

Second, cultural attribution is inherently underdetermined from image evidence alone. Many images require contextual priors that are unavailable from pixels.

Third, retrieval quality remains an open problem. Manual inspection revealed that many medium-confidence retrievals matched scene categories rather than specific cultural artifacts. Concept-rate statistics may therefore reflect a mixture of genuine grounding and retrieval-induced label copying.

Fourth, our human evaluation was conducted by researchers who were neither community members nor speakers of the target languages. Cultural judgments should therefore be interpreted as measurements of specificity rather than definitive correctness.

Fifth, gold captions frequently contain nonvisual contextual knowledge (§6), which imposes a structural ceiling on image-only captioning systems and complicates reference-based evaluation.

Finally, we did not conduct target-language human evaluation due to the absence of available native-speaker annotators during the project period.

Ethics Statement

Our goal is to support Indigenous-language technology without replacing community expertise. Generated captions are not authoritative cultural interpretations and should not be deployed in community-facing contexts without review by native and heritage speakers.

We adopt a conservative policy that prohibits naming photographed individuals unless their identity is explicitly present in visible image text. This decision follows observed cases where retrieval systems incorrectly attached real cultural figures to unrelated individuals.

All datasets and retrieval sources were used according to their published licenses. Image retrieval collections were harvested only from appropriately licensed Wikimedia Commons content, and translation was performed through a paid-tier API configuration compatible with CICIL licensing requirements. The Maya retrieval bank uses the YUA-ES Communicative Contexts Corpus (Molina-Villegas et al., 2026), cited per its release requirements.

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